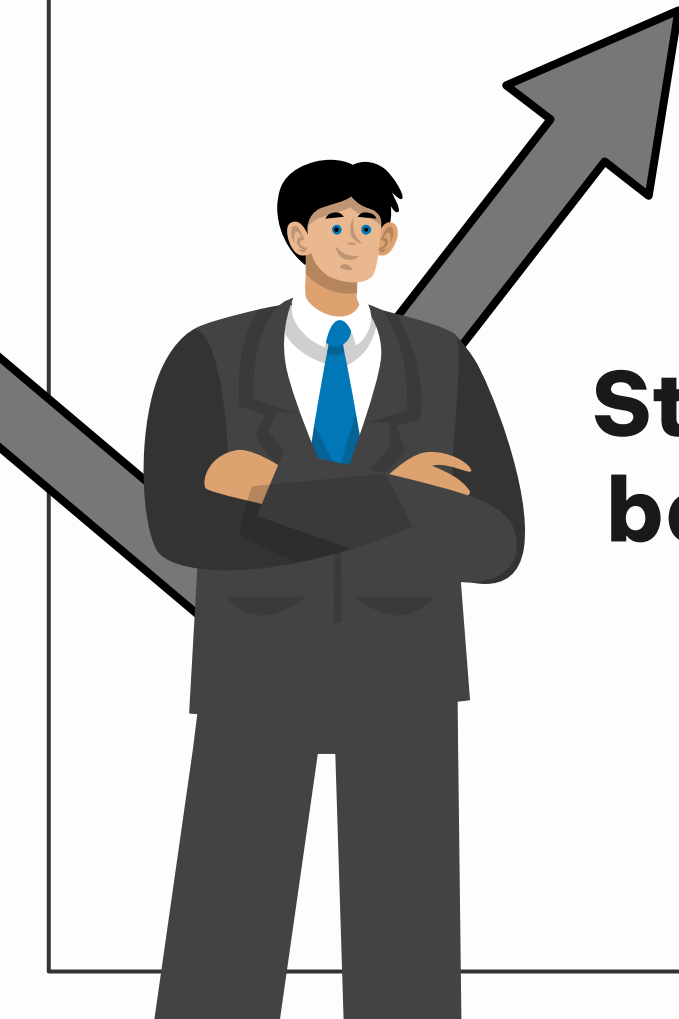




Stock Market Predictions based on US Presidential Results

Politics, Stock Prices, & Predictive Power

Project Goal

This project aims to develop a predictive algorithm to forecast stock performance by analyzing the relationship between stock trends and U.S. presidential election results from 2000 to 2024. Understanding this relationship can provide valuable insights into how politics influences different stock sectors.

Principal Goals:

- Analyze past election patterns and their impact on market sentiment.
- Examine sectoral stock performance under various political regimes.
- Develop a predictive algorithm to forecast stock price changes over the course of a presidential term based on election outcomes.
- Assess risks and constraints, including

Problem Statement and Justification:

U.S. presidential elections impact the stock market, but no model reliably predicts sector-specific performance. Understanding these patterns can aid investors' decisions.

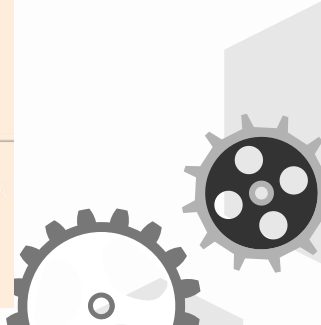
Key Considerations:

- **Sector-Specific Variability:** Industries like tech, healthcare, and energy react differently under varying political leadership, requiring a nuanced analysis.
- **Machine Learning Utility:** ML captures complex patterns using election and economic data, improving prediction accuracy.
- **Risk Management:** Managing data bias, market volatility, and overfitting ensures reliable, real-world results.



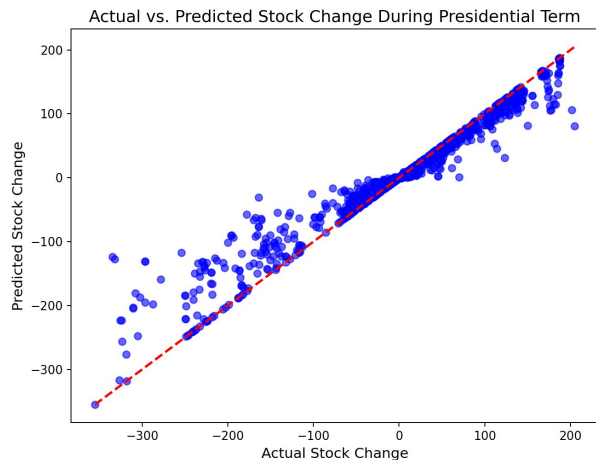
Understanding Our Data

Dataset	Source	Description	Time Frame	Rows	Sample Columns	Data Type Examples
U.S. Presidential Election Data	270toWin 2000 Election Results	Contains U.S. presidential election results, including popular vote percentages, electoral votes, and winning party per state.	2000 to 2024 elections (every 4 years)	~6,300 (daily records for a U.S. index from 2000–2024)	State, Democratic Vote %, Republican Vote %, Electoral Votes, Winning Party	Categorical: State, Winning Party Numeric: Vote %, Electoral Votes
Stock Market Data	Kaggle - World Stock Prices	Daily closing prices for global stock indexes, used to examine U.S. stock market movement around elections.	2000-2024 (daily frequency)	~1,000,000+ rows (subset used for U.S. indices)	Date, Ticker, Open, Close, Volume	Date: Date Numeric: Open, Close, Volume Text: Ticker
Voter Demographics Data	U.S. Census - Voting Historical Time Series (Table A-9)	Provides voter turnout rates by age group and gender across federal elections. Provides insight into who is voting and how their preferences might shape election outcomes.	2000–2020 presidential elections	~50 rows (aggregated by year and demographic)	Year, Age Group, Turnout Rate (%), Gender	Categorical: Age Group, Gender Numeric: Turnout Rate (%) Date/Integer: Year
Financial Metrics	Investopedia	Contains key macroeconomic indicators recorded during each presidential election cycle.	2000 to 2024 (election years only)	7 (one per election year)	Unemployment Rate, Interest Rate, Inflation Rate	Numeric: Unemployment Rate (%), Interest Rate (%), Inflation Rate (%)



Exploring Predictive Possibilities

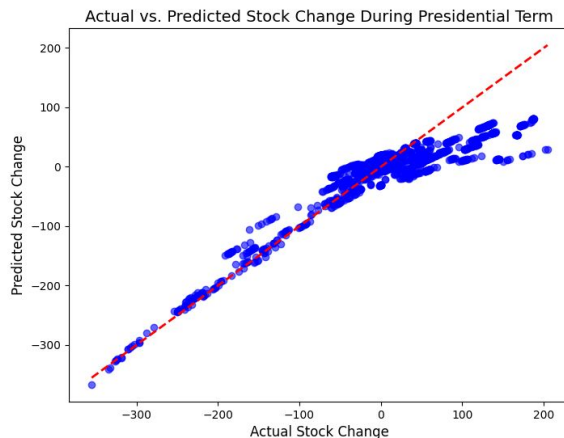
KNN Model



MSE: 271.72 RMSE: 16.48 R^2 : 0.93

- KNN has the ability to capture local patterns.
- The R^2 value and RMSE shows strong predictive power.
- KNN can be prone to overfitting when the number of neighbours is small, leading to high variance and poor generalization of unseen data

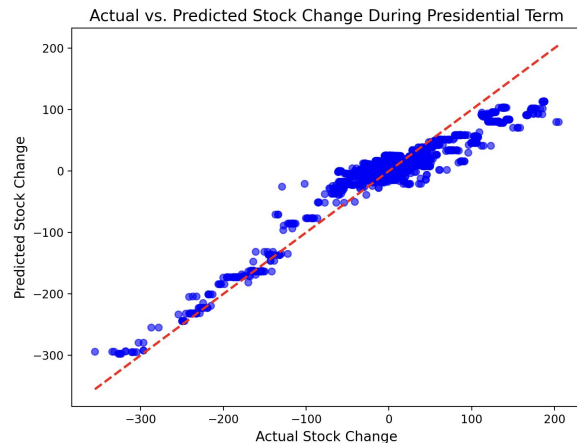
SVM



MSE: 846.79 RMSE: 29.10 R^2 : 0.78

- SVM is able to handle non-linear relationships with the use of kernels.
- The lower R^2 value suggests a higher bias, implying that model might be too simple for the dataset.
- SVM's sensitivity to noise can lead to potential overfitting with carefully tuned hyperparameters

Gradient Boosting



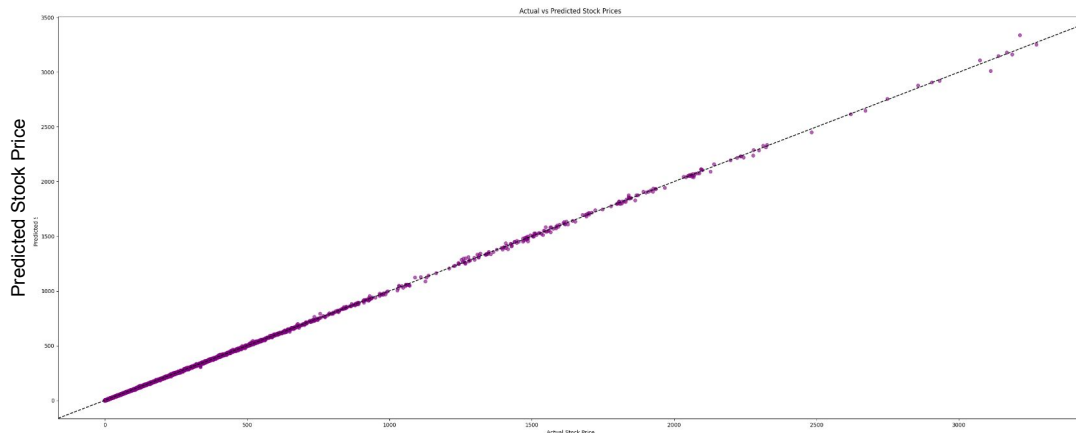
MSE: 478.78 RMSE: 21.88 R^2 : 0.88

- Gradient Boosting is an ensemble method that iteratively corrects its own errors.
- It captures non-linear relationships well, but there is a risk of overfitting due to its iterative nature

Getting Started: Initial Random Forest Model

Random Forest

Actual vs Predicted Stock Price



Actual Stock Price

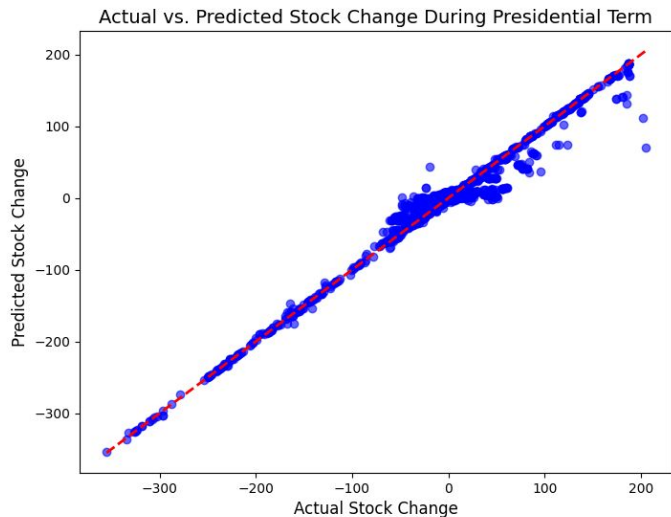
	MSE	MAE	R ²
Results	3.30	0.56	1.00

Actual Price	Predicted Price
14.310377	14.137915
3.340000	3.325643
295.663269	294.613911
34.121315	34.034011
41.480328	41.734946

- Random Forest combines multiple decision trees to model complex relationships and reduce variance
- Initial model showed perfect metrics:
 - Perfect results indicated overfitting, not true learning
- Financial features with large magnitudes (e.g., stock price, interest rate) dominated predictions
- Model likely memorized training data, failing to generalize
- Introduced **cross-validation** and **hyperparameter tuning** → To improve generalization and ensure real-world forecasting reliability

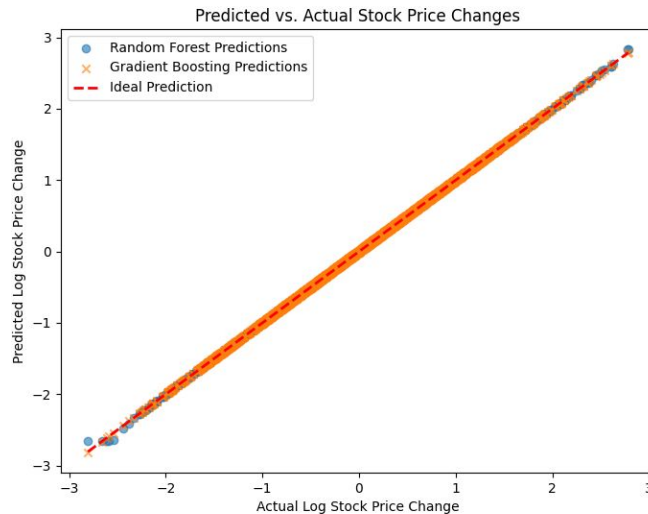
Overfitting - When Models are TOO Good

Random Forest: k=3 Cross-Validation



	K-Fold Cross Validation	GridSearchCV
MSE	118.54	0.00
RMSE	10.89	
R²	0.97	1.00
Gradient Boosting R²		1.00

Random Forest: GridSearchCV on Aggressive Features



Best RF Parameters:

- bootstrap: True
- max_depth: 12
- max_features: sqrt
- min_samples_leaf: 10
- min_samples_split: 15
- n_estimators: 250

Initial Model: RF with k=3 for faster development.

- Achieved R² of 0.97 with low error, raising concerns of overfitting.

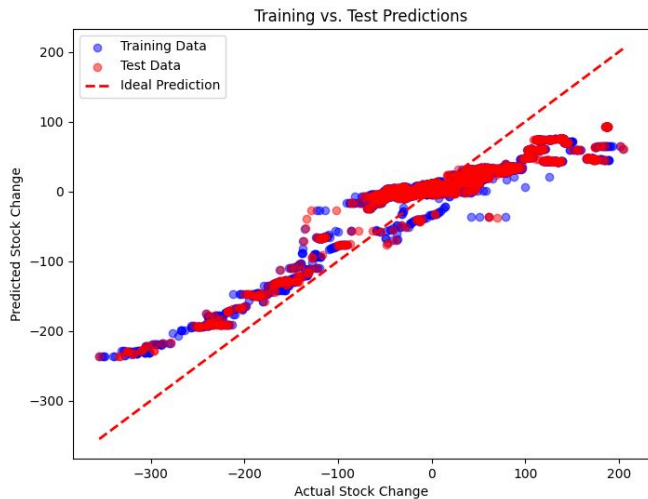
Model Adjustments: Applied GridSearchCV and shifted target to percentage change (log-scaled); encoded categorical features.

- Despite 5-fold validation, both Random Forest and Gradient Boosting models achieved perfect R² (1.00), indicating overfitting.

Conclusion: Model's “perfect” performance is unreliable for forecasting due to overfitting and lack of generalizability.

Fine-Tuning the Random Forest Model

Random Forest: Hyperparam Tuning



- RandomizedSearchCV (10 iterations) for efficient hyperparameter tuning
- Regularized parameters to prevent overfitting:
 - Shallow trees (low max depth)
 - Higher min samples per split/leaf
 - Limited feature usage per split
 - Bootstrap toggled to assess ensemble variability
- Separated preprocessing with ColumnTransformer (MinMax + OneHotEncoder)
- Evaluated with 5-fold CV (k=5) – stable, low variance, and efficient

Results:

- $CV R^2 = 0.778 \pm 0.011$
- $Test R^2 = 0.779$, $Train R^2 = 0.791$
- $RMSE = 29.29$ (realistic error for stock data)
- Best balance of accuracy, generalization, and interpretability

Real World Test : McDonald's (NYSE: MCD) and the 1996 Election

Inputted Values :

Opening Price Before Election	\$11.92	Election Year Inflation Rate	5.25%
Election Year Unemployment	5.4%	Party	Democratic (D)
Election Year Interest Rate	5.3%	Industry	Food

Model Predicted Value at the End of Term: \$15.42

Actual Value of MCD at End of Term: \$17.39

- Accurate Directional Prediction: The model successfully forecasted the stock's price, predicting an increase with a conservative estimate

Results and Comparative Analysis

Model	R ² Score	RMSE	MSE	Notes
KNN	0.93	16.48	271.72	Strong performance; underestimates extreme changes
SVM	0.78	29.10	846.79	Weaker generalization; struggles with sharp trends
Gradient Boosting	0.88	21.88	478.78	Balanced accuracy; good at capturing non-linear patterns
Random Forest (All Features)	0.999	3.30	3.30	Extremely accurate, likely overfitting due to strong financial bias
Random Forest (K-Fold, k=3)	0.97	10.89	118.54	Best generalization; consistent and robust performance
Random Forest (Hyperparameter Tuning, k=5)	0.78	29.29	857.87	Consistent train/test scores, but less accurate than K-Fold model

Final Thoughts

Recommended Model : Random Forest with Hyperparameter Tuning (k=5)

Emerged as the most reliable and balanced model overall!

Final Performance:

- Indicates low variance and strong real-world potential

R² (Train)	0.791
R² (Test)	0.779
Cross-Validation R²	0.778 ± 0.011

What Made It Effective:

- Tuned key hyperparameters (tree depth, split size, feature limit)
- Reduced overfitting while retaining predictive strength
- Generalized well across multiple political and economic conditions

Why It Outperformed the Others:

- Other models (e.g., KNN, RF with k=3) had higher R², but showed overfitting or unstable performance
- This model offered the best tradeoff between:
 - Accuracy
 - Generalization
 - Stability

Conclusion:

- Despite not having the highest R², the tuned Random Forest (k=5) is our top choice for deployment, offering the most stable, realistic, and interpretable predictions of stock behavior in response to presidential elections.

Bibliography

Bureau, US Census. "Voting and Registration in the Elections." Census.gov, Apr. 2023, www.census.gov/data/tables/time-series/demo/voting-and-registration/p20-586.html.

"K-Nearest Neighbours Algorithm." Data Science Stack Exchange, 24 Apr. 2019, datascience.stackexchange.com/questions/50847/k-nearest-neighbours-algorithm-explanation-needed.

Koehrsen, Will. "Random Forest in Python | towards Data Science." Towards Data Science, 27 Dec. 2017, towardsdatascience.com/random-forest-in-python-24d0893d51c0/.

"Presidential Election of 2000s - 270toWin." 270toWin.com, 2024, 270towin.com/2000-election.

Reiff, Nathan. "Historical US Unemployment Rate by Year." Investopedia, 17 Oct. 2023, www.investopedia.com/historical-us-unemployment-rate-by-year-7495494.

"U.S. Inflation Rate by Year: 1929-2023." Investopedia, 17 May 2023, www.investopedia.com/inflation-rate-by-year-7253832.

"World Stock Prices (Daily Updating)." Wwww.kaggle.com, www.kaggle.com/datasets/nelgiriyeewithana/world-stock-prices-daily-updating.

Repository Including Datasets:

https://github.com/smrithishiv/mse446_mlproject

